

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Selected Professional Experience

- 08/2018-8/2021 Project Scientist, NASA SnowEx Mission
- 08/2016-07/2018 Co-lead of Field Campaign, NASA SnowEx 2017 Mission
- 08/2013-present Associate Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
- 07/2007-05/2020 Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing.
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
Computational statistics, geostatistics, compressed sensing, instrumentation development

Selected Honors and Awards

- 2020 BSU Faculty Excellence Award
- 2019 NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
- 2019 2018 WRR Editors' Choice Award: Hedrick et al., 2018
- 2018 NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
- 2012 Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
- 2010-13 New Investigator Program Award, NASA, 1st in Idaho history
- 2010 Young Scientist Award, Cryosphere Section, American Geophysical Union
- 2009 Top Reviewer Award, Cold Regions Science and Technology (CRST)
- 2008 U.S. Young Scientist Award, International Union of Radio Science (URSI)

Selected Recent Peer-Reviewed Publications

[total=72, h-index=29, i10-index=58, citations=2742]

- 2022 Lievens, H., Brangers, I., **H.-P. Marshall**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc-16-159-2022
- Lund, J., Forster, R., Deeb, E., Liston, G., Skiles, M., and **H.P. Marshall** (2022). Interpreting sentinel-1 sar backscatter signals of snowpack surface melt/freeze, warming, and ripening, through field measurements and physically-based snowmodel. *Remote Sensing*, 14:4002, <https://doi.org/10.3390/rs14164002>
- McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **H.P. Marshall** (2022). A time-series of snow density and snow water equivalent observations derived from the integration of gpr and uav sfm observations. *Frontiers in Remote Sensing*, 3:886747, doi:10.3389/frsen.2022.886747
- Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **H.-P. Marshall**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2022). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere*, 16(9):3531–3573
- Lemmetyinen, J., Cohen, J., Kontu, A., Hannula, J. V. H., Merkouriadi, I., Scheiblaue, S., Rott, H., Nagler, T., Ripper, E., Elder, K., **H.P. Marshall**, Fromm, R., Adams, M., Derksen, C., King, J., Meta, A., Coccia, A., Rutter, N., Sandells, M., Macelloni, G., Santi, E., Leduc-Leballeur, M., Essery, R., Menard, C., and Kern, M. (2022). Airborne snowsar data at x- and ku- bands over boreal forest, alpine and tundra snow cover. *Earth Syst. Sci. Data*, 14(9):3915–3945
- 2021 **H.-P. Marshall**, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852
- Meehan, T. G., **H.P. Marshall**, Bradford, J. H., Hawley, R. L., Overly, T. B., Lewis, G., Graeter, K., Osterburg, E., and McCarthy, F. (2021). Historical surface mass balance reconstruction 1984-2017 from greentracs multi-offset ground-penetrating radar. *Journal of Glaciology*, 67(262):219–228, doi:10.1017/jog.2020.91
- Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. (2020). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318, doi:10.3389/feart.2019.00318
- 2019 Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12